CHM111 Practice Exam on chapter(s): 5, 6

Name:____

These questions assess your understanding of the material from Chapters 5, 6 of Silberberg. Please write your name on the sheets of paper that you use and show all your work.

1. A chemist carefully measures the amount of heat needed to raise the temperature of a 1.54-kg sample of a pure substance from 4.1°C to 9.49°C. The experiment shows that 2.3 kJ of heat is needed. What can the chemist report for the specific heat capacity of the substance?

   ANSWER: ____________________

2. The vapor pressure of a particular compound at 66°C is 320. torr. Calculate the vapor pressure in atm.

   ANSWER: ____________________

3. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 125.8 MPa. The student's design calls for an observation port on the side of the vessel. The bolts that hold the cover of this port onto the vessel can safely withstand a force of 3.24 MN. Calculate the maximum safe diameter of the port.

   ANSWER: ____________________

4. For many purposes, we can treat propane (C_3H_8) as an ideal gas at temperatures above its boiling point of -43.°C (230.15 K). Suppose the pressure on a 543.-mL sample of propane gas at 290. K is reduced to 32% of its initial value. Is it possible to maintain the volume of the propane gas during this pressure reduction by adjusting the gas's temperature? If you believe it is, calculate the temperature of the gas after its pressure is reduced. Round your answer to the nearest °C.

   ANSWER: ____________________

5. Carbon disulfide gas and oxygen gas react to form sulfur dioxide gas and carbon dioxide gas. What volume of carbon dioxide would be produced by this reaction if 3.3 cm^3 of carbon disulfide were consumed?

   ANSWER: ____________________

6. The vapor pressure of a particular compound at 74°C is 349. torr. Calculate the vapor pressure in mmHg.

   ANSWER: ____________________

7. A cylinder measuring 12.6-cm wide and 5.3-cm tall is filled with gas. The piston is pushed down with a steady 54.2-N force. Calculate the pressure of the gas inside the cylinder. Calculate the pressure of the gas inside the cylinder. Report your answer in units of kilopascals and with the correct number of significant figures.

   ANSWER: ____________________

8. Tyler is preparing a deep-sea submersible for a dive. The sub stores breathing air under high pressure in a spherical air tank that measures 86.4-cm wide. He estimates that he will need 6.57×10^3 L of air for the dive. Calculate the pressure to which this volume of air must be compressed in order to fit into the air tank. Report your answer in atmospheres.

   ANSWER: ____________________

9. A gas-storage cylinder in an ordinary chemical laboratory measures 6.0-cm wide and 25.0-cm tall. Its label reads "Contents: Nitrogen gas" and "Pressure: 6.34 atm". If the cylinder is opened and the gas is allowed to escape into a large, empty plastic bag, what will the final volume of nitrogen gas be, including what is collected in the plastic bag and what is left over in the cylinder? Report your answer in liters.

   ANSWER: ____________________

10. For many purposes, we can treat propane (C_3H_8) as an ideal gas at temperatures above its boiling point of -42.°C. Suppose the temperature of a sample of propane gas is lowered from 1.1°C to -37.0°C as its pressure is allowed to change. If the initial pressure was 5.41 atm and the volume increased by 58%, what is the final pressure?

    ANSWER: ____________________

11. A reaction between liquid reactants takes place at 13.°C in a sealed, evacuated vessel with a measured volume of 25.8 L. Measurements taken after the reaction show that 4.15 g of gaseous carbon dioxide was produced. Calculate the pressure of carbon dioxide gas in the reaction vessel after the reaction. You may ignore the volume of the liquid reactants.

    ANSWER: ____________________

12. A particular bowl of yogurt has an energy content of 58. kcal. What is this in kilojoules?

    ANSWER: ____________________

13. The vapor pressure of a particular compound at 7°C is 0.0104 atm. Calculate the vapor pressure in kPa.

ANSWER: ________________________

14. Kevin puts a 0.23-kg aluminum pot on the stove and fills it with 0.69 liters of water. Both the pot and the water are initially at 0.28°C. How much heat is required to bring them both to 53.4°C? The specific heat value of water is 4186. J/kg°C and for aluminum it is 900. J/kg°C.

ANSWER: ________________________

15. A mixture of sulfur tetrafluoride and chlorine gas is expanded from a volume of 110 L to a volume of 120 L while the pressure is held constant at 58.6 atm. Calculate the work done on the gas mixture and report your answer to 3 significant figures.

ANSWER: ________________________

16. An arctic weather balloon is filled with 2.25 L of helium gas inside a shed. The temperature inside the shed is 11. °C. The balloon is then taken outside, where the temperature is -27. °C. Calculate the new volume of the balloon, assuming that the pressure on the balloon stays constant at 1 atm.

ANSWER: ________________________

17. Carbon disulfide gas and oxygen gas react to form sulfur dioxide gas and carbon dioxide gas. What volume of sulfur dioxide would be produced by this reaction if 6.05 cm^3 of carbon disulfide were consumed?

ANSWER: ________________________

18. Carbon disulfide gas and oxygen gas react to form sulfur dioxide gas and carbon dioxide gas. What volume of sulfur dioxide would be produced by this reaction if 10.2 cm^3 of oxygen gas were consumed?

ANSWER: ________________________

19. Measurements show that the enthalpy of a mixture of gaseous reactants decreases by 153.5 kJ during a certain chemical reaction, which is carried out at a constant pressure. Furthermore, by carefully monitoring the volume change, it is determined that -154.5 kJ of work is done on the mixture during the reaction. Calculate the change in energy of the gas mixture during the reaction and report whether is it endothermic, exothermic, or neither.

ANSWER: ________________________

20. Kevin puts a 0.23-kg aluminum pot on the stove and fills it with 0.69 liters of water. Both the pot and the water are initially at 0.28°C. How much heat is required to bring them both to 53.4°C? The specific heat value of water is 4186. J/kg°C and for aluminum it is 900. J/kg°C.

ANSWER: ________________________

21. Hydrogen gas and nitrogen gas react to form ammonia gas. What volume of ammonia would be produced by this reaction if 3.11 mL of nitrogen were consumed?

ANSWER: ________________________

22. A 12.6-L tank at 110°C is filled with 7.67 g of chlorine (Cl_2) and 13.8 g of methane (CH_4). You can assume both gases behave as ideal gases under these conditions. Calculate the mole fraction of methane in this mixture.

ANSWER: ________________________

23. A 7.53-L tank at 110°C is filled with 9.1 g of propylene (C_3H_6) and 11. g of chlorine (Cl_2). You can assume both gases behave as ideal gases under these conditions. Calculate the partial pressure of propylene in this mixture.

ANSWER: ________________________

24. A sample of an unknown compound is vaporized at 130.°C. At a pressure of 1.31 atm, the gas produced has a volume of 920 mL and a mass of 1.38 g. Assuming the gas behaves as an ideal gas under these conditions, calculate the molar mass of the compound.

ANSWER: ________________________

25. Carbon dioxide gas is collected at 3.5°C in an evacuated flask with a measured volume of 5.9 L. When all the gas has been collected, the pressure in the flask is measured to be 0.583 atm. Calculate the amount of carbon dioxide gas that was collected. Report your answer in moles.

ANSWER: ________________________

26. Suppose that there is heat transfer of 37 J to a system, while the system does 31 J of work. Later, there is heat transfer of 54 J out of the system while 41 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: ________________________

27. A cylinder is filled with 8.0 L of gas and a piston is put into it. The initial pressure of the gas is measured to be 220 kPa. The piston is then pushed down to compress the gas until its volume is 2.07 L. Calculate the pressure of the compressed gas.

ANSWER: ______________________

28. The vapor pressure of a particular compound at 9°C is 0.5190 kPa. Calculate the vapor pressure in atm.

ANSWER: ______________________

29. The partial pressure of nitrogen in the atmosphere at a certain elevation is 525.1 mmHg. Calculate the partial pressure in atm.

ANSWER: ______________________

30. Calculate to three significant digits the density of ethane gas (C_2H_6) at 10.2°C and 1.10 atm. Assume that the gas behaves as an ideal gas under these conditions.

ANSWER: ______________________

31. A 11.7-L tank at 72.0°C is filled with 7.5 g of methane (CH_4) and 10. g of carbon dioxide (CO_2). You can assume both gases behave as ideal gases under these conditions. Calculate the mole fraction of methane in this mixture.

ANSWER: ______________________

32. Hydrogen gas and nitrogen gas react to form ammonia gas. What volume of ammonia would be produced by this reaction if 4.24 mL of hydrogen were consumed?

ANSWER: ______________________

33. The partial pressure of nitrogen in the atmosphere at a certain elevation is 626. mmHg. Calculate the partial pressure in torr.

ANSWER: ______________________

34. A cylinder measuring 7.1-cm wide and 12.-cm tall is filled with gas. The piston is pushed down with a steady force. Calculate the force on the piston if the pressure in the gas is measured to be 19. kPa. Report your answer in units of newtons and with the correct number of significant figures.

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